Guide: How to Crimp CAT6 Cables with RJ45 Connectors for Different Purposes

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Part 1: Crimping CAT6 for Power Only (Passive PoE)

Purpose: Use only certain pairs to carry power (not data), typically for passive PoE applications like powering access points or cameras.

Required Tools & Materials:

- CAT6 cable
- RJ45 connectors
- Crimping tool
- Wire stripper

Wiring Scheme (Power Only - No Data):

Use these pairs:

- Pair 4: Blue (Pin 4)
- Pair 5: Blue/White (Pin 5)
- Pair 7: Brown/White (Pin 7)
- Pair 8: Brown (Pin 8)

Steps:

1. Strip off about 1 inch of the outer CAT6 jacket.
2. Untwist and align the needed wires (blue, blue/white, brown, brown/white).
3. Trim to even length.

4. Insert only the 4 selected wires into the RJ45 connector matching the pin positions.

5. Leave data pins (1,2,3,6) empty.

6. Crimp the connector securely.

Connecting to Power Adapter Directly:

1. Identify the positive and negative legs of your power adapter (use a multimeter if unsure).

2. Use a wire stripper to expose 12 cm of the copper from the relevant CAT6 wires:

   - Connect Pin 4 (Blue) and Pin 5 (Blue/White) together for positive (+).

   - Connect Pin 7 (Brown/White) and Pin 8 (Brown) together for negative (-).

3. Twist the positive wires together and solder or tightly twist to the positive leg of the adapter.

4. Do the same for the negative wires and join them to the negative leg of the adapter.

5. Secure the connection with insulation tape or heat shrink tubing.

Note: The other end must match this configuration or use a power injector/splitter accordingly.

Part 2: Crimping CAT6 for Internet Only (Data Only)

Purpose: To transmit data (Internet) without power.

Wiring Scheme (T568B Standard):

- Pin 1: Orange/White

- Pin 2: Orange

- Pin 3: Green/White

- Pin 6: Green

Steps:

1. Strip about 1 inch of jacket.
2. Untwist all pairs, then select the required data pairs (pins 1, 2, 3, 6).
3. Arrange in T568B sequence.
4. Cut ends evenly and insert into RJ45.
5. Crimp firmly.

Note: Pins 4, 5, 7, 8 will be unused and can be left out or inserted without connection.

Part 3: Crimping CAT6 for Internet + Power (Split for Passive PoE)

Purpose: One cable carries both data and power. Data uses standard pins, power uses unused pairs.

Wiring Scheme (T568B + Power):

- Data (T568B):
  - Pin 1: Orange/White
  - Pin 2: Orange
  - Pin 3: Green/White
  - Pin 6: Green
- Power (Passive PoE):
  - Pin 4: Blue (+)
  - Pin 5: Blue/White (+)

- Pin 7: Brown/White (-)
- Pin 8: Brown (-)

Steps:

1. Strip the CAT6 cable jacket (1 inch).
2. Untwist and align all 8 wires.
3. Arrange in full T568B standard (with PoE pairs intact).
4. Insert all 8 wires into RJ45 connector.
5. Crimp firmly.
6. At the device/injector end, break out pins 4,5 and 7,8 for power tap (using a PoE splitter or injector).

Caution: Passive PoE is not 802.3af/at compliant and may damage equipment if voltage/current is not regulated.

Summary Table

Use Case	RJ45 Pins Used	Description
Power Only	4,5,7,8	For passive PoE power delivery only
Internet Only	1,2,3,6	Standard Ethernet (T568B)
Internet + Power	1,2,3,6 + 4,5,7,8	Combined data and power in one cable

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End of Guide